SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: **Total Marks: 10**

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

```
A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
```

```
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>
```

Snippet 2: Form Submission Methods

```
A. GET
B. POST
C. PUT
D. DELETE
```

Snippet 3: Navigation Elements

```
A. <div>
B. <nav>
C. <section>
D. <header>
```

Snippet 4: XML Syntax

```
A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note
```

Snippet 5: Web Communication Protocols

```
A. FTP
B. HTTP
C. SMTP
D. SNMP
```

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

I ***Narjala Saptha Nageswar - 192372001*** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *N Saptha Nageswar*

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANSWERS

Internet Programming (CSA43) – Assessment Tool 3: Web Development Standards Implementation

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1. Video embedding

Answer: Option B – <video src="video.mp4" controls></video>

Justification:

- **<video>** is the standard HTML5 element for embedding video, so the browser plays it natively without any plug-in such as Flash.
- The **src** attribute gives the file path, and the **controls** attribute shows play, pause, volume and fullscreen buttons to the visitor.
- Options A (<media>), C (<movie>) and D (<vid>) are not valid HTML elements; browsers ignore them and nothing is displayed.

```
<video src="video.mp4" controls width="640">
  Your browser does not support the video tag.
</video>
```

2. HTTP method for form submission

Answer: Option B – POST

Justification:

- POST sends the form data in the **body** of the HTTP request, so values such as passwords and emails do not appear in the URL, browser history or server logs.
- GET appends data to the URL as a query string, which is visible and unsafe for registration data, and it is limited in length.
- PUT is used to replace/update an existing resource and DELETE is used to remove a resource; neither is meant for submitting a new registration form.
- POST is also used for creating new records, which matches user registration. It should be combined with HTTPS for real security.

```
<form action="/register" method="POST">
  <input type="text" name="username">
  <input type="password" name="password">
  <button type="submit">Register</button>
</form>
```

3. Semantic element for navigation

Answer: Option B – <nav>

Justification:

- **<nav>** is the HTML5 semantic element that marks a block of major navigation links, so its purpose is clear from the tag itself.
- Screen readers and other assistive tools can jump straight to it, and search engines understand the site structure better (accessibility and SEO).
- has no meaning, <section> groups thematic content, and <header> holds introductory content; none of these specifically identifies navigation.

```
<nav>
  <ul>
    <li><a href="index.html">Home</a></li>
    <li><a href="about.html">About</a></li>
    <li><a href="contact.html">Contact</a></li>
  </ul>
</nav>
```

4. Correct XML syntax

Answer: Option B – <note><to>User</to><from>Admin</from></note>

Why it is correct: every opening tag has a matching closing tag, the tags are properly nested, and there is a single root element (<note>), so the document is well-formed.

Errors in the other options:

- **A:** The root element <note> is never closed; the closing </note> tag is missing.
- **C:** <to> is closed with </from>. Tags are mismatched and improperly nested, and XML is case- and name-sensitive.
- **D:** The closing tag is incomplete (</note without the >), so the markup is malformed.

5. Protocol for web communication

Answer: Option B – HTTP

Justification:

- HTTP (HyperText Transfer Protocol) is the application-layer protocol used between web browsers and web servers to request and deliver web pages, images, video and other content, using methods such as GET and POST. Its secure version, HTTPS, adds TLS encryption.
- FTP is meant for file transfer, SMTP for sending email, and SNMP for managing network devices; none of them is used for delivering web content.